

REMOTE SOURCE

888-769-8806

inquiries@remotesource.net

www.remotesource.net

Current State of the Industry

Our Earth is not made to absorb all the toxic waste that humans create.

The world consumes 200 million remote controls every year, each using an average of 2 batteries.

The average battery life for a remote control is 6 months.

This means an estimated 800 million batteries are used and discarded every year!

We all have to do our part to create a more sustainable future.

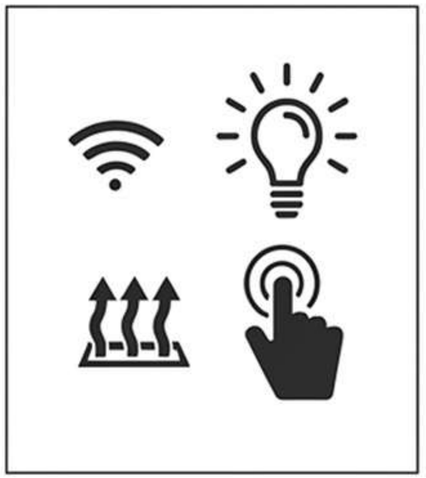


Energy Harvesting is changing the POWER equation

Extreme Low-power Bluetooth®

+

Energy Harvesting



Energy Storage

+ Capacitor = Battery Free

+ Standard Battery = Extended Battery Life

+ Rechargeable Battery = Batteries Last Forever

Battery Free Living

- ✓ Power without diposable battries
- ✓ Seamless OCA integration with a hidden/invisible PV panel
- ✓ Forget about battery replacement
- ✓ Green energy for everyday use
- ✓ The Market's most advanced LCA



Exceptional sustainability profile



- Significantly lower CO_2 impact across product lifecycle compared to legacy solutions
- Less material and energy used in production

Unique integration capabilities



- Exceptionally thin (0.2 mm) and flexible modules enable easy integration
- Additional features such as MultiCell, functional surfaces, bi-facial.

Solar RCU PV panel alternatives

Various photovoltaic panels are available for different scenarios and usage intensities.

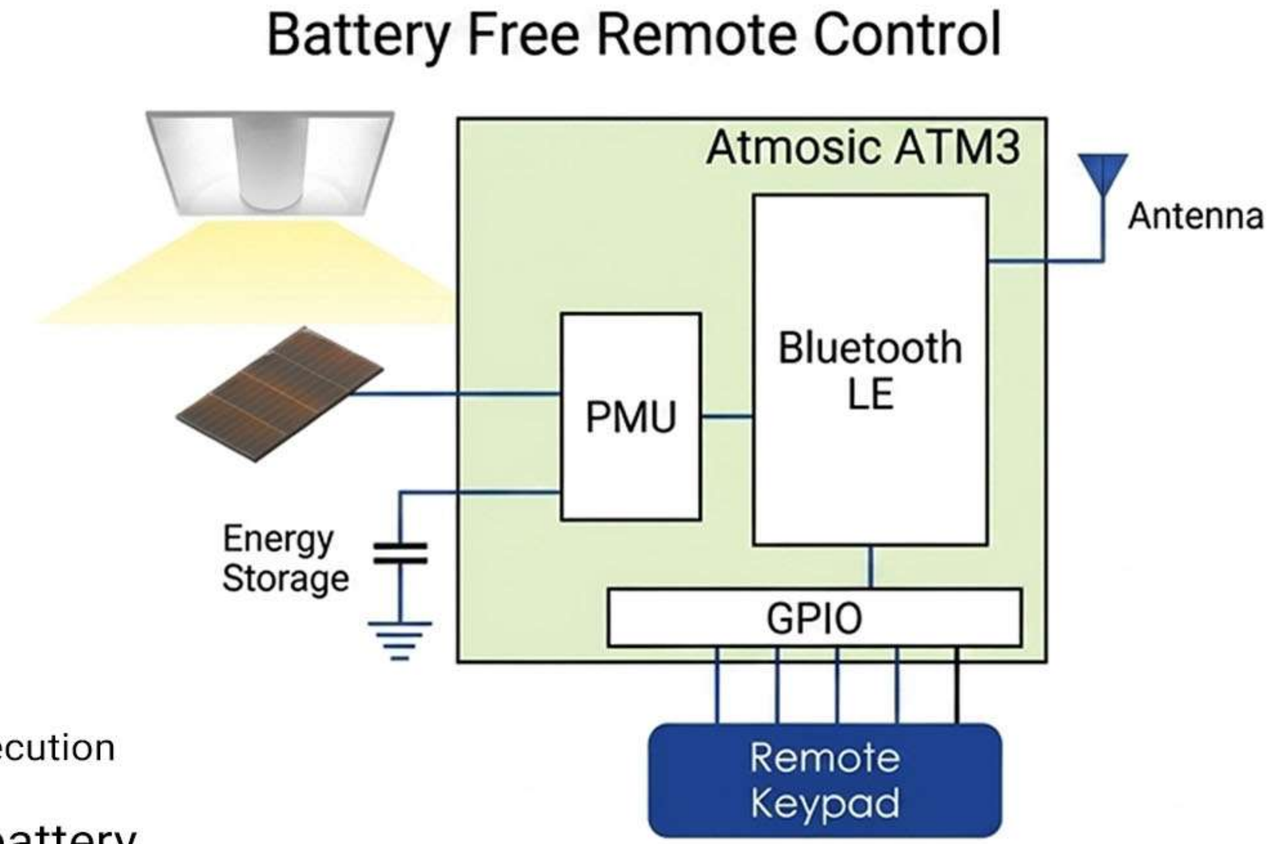
The LOWER the Relative cost comparison, the BETTER the cost-efficiency.

Table 1. PV panels technologies relative cost comparison

PV panel technology	Vendor	Power density uW/cm2 @100 lux Before integration	Power density uW/cm2 @200 lux Before integration	Relative cost comparison, %
DSSC (Dye-Sensitized Solar Cell)	Ambient Photonics, USA	TBD	14.00	121%
a-Si(Amorphic Silicon)	Trony, China	1.12	4.87	100%
PowerFoyle (DSSC) Outdoor	Exeger, Sweden	0.19	9.39	82%
Printed organic solar cells	Epishine, Sweden	1.92	7.83	65%
Pirovskite panels	SAULE, Poland	1.70	7.94	58%

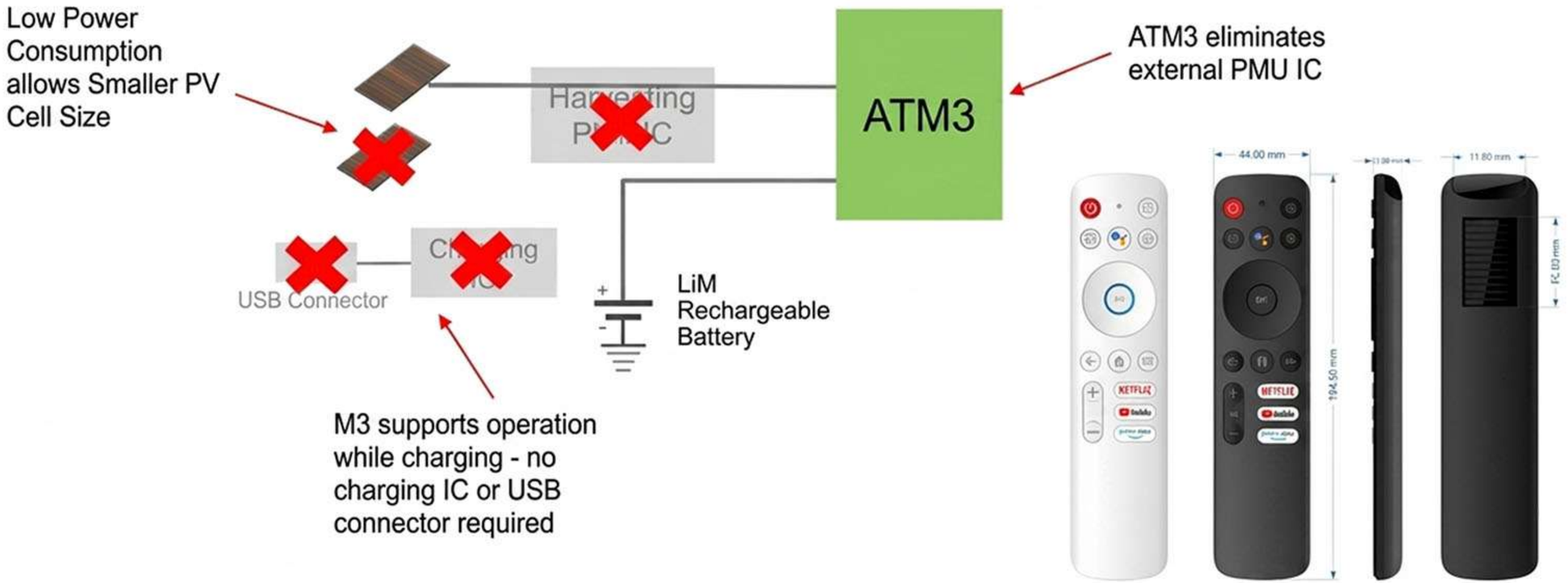
Remote Source's Single IC solution for BLE+PM

- ✓ Integrated ultra low power BLE and Power Management Unit (PMU)
- ✓ Direct connection to PV cell
- ✓ Maximizes harvesting efficiency
- ✓ Reduces overall BOM
- ✓ PMU increases harvesting efficiency
  - ✓ Directly manages PV cell
  - ✓ Delivers energy required for BLE operation
  - ✓ Stores excess energy not required for immediate execution
- ✓ Energy storage in capacitor or rechargeable battery



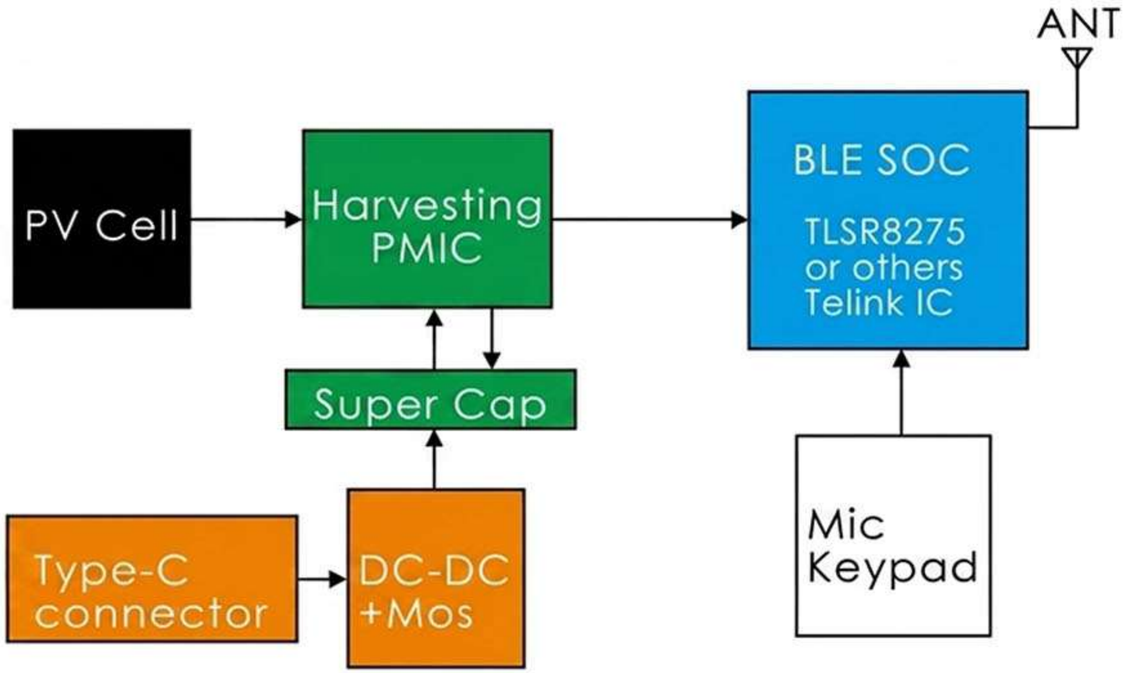
Atmosic ATM3 For MTK mainboard ATV GTV and All AOSP mainboard.

Energy Harvesting & Electronics Optimizing



Remote Source's 2 ICs Energy Harvesting Solution 1

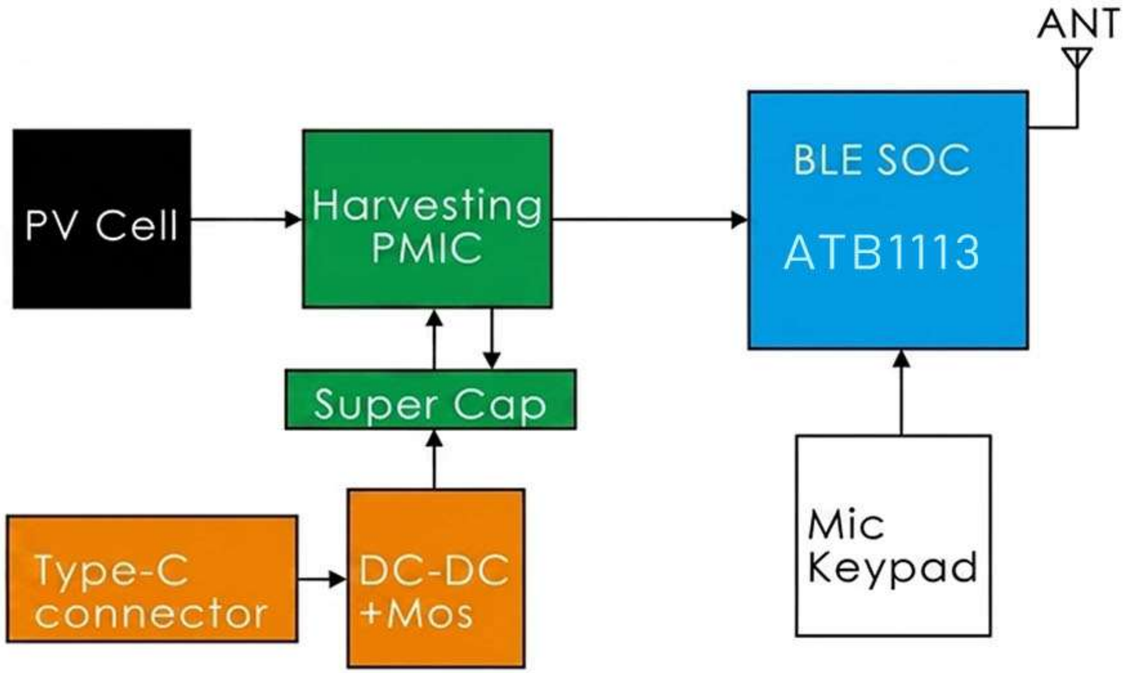
- ✓ Low power BLE SOC
- ✓ Can adapt to various well-known PV cell brands
- ✓ Can efficiently harvests energy in Low-light
- ✓ Delivers the energy required for BLE operation
- ✓ Battery Free(Energy stored in Super Cap)
- ✓ GTV/ATV/AOSP



TLSR8275 For MTK mainboard ATV GTV and All AOSP mainboard.

Remote Source's 2 ICs Energy Harvesting Solution 2

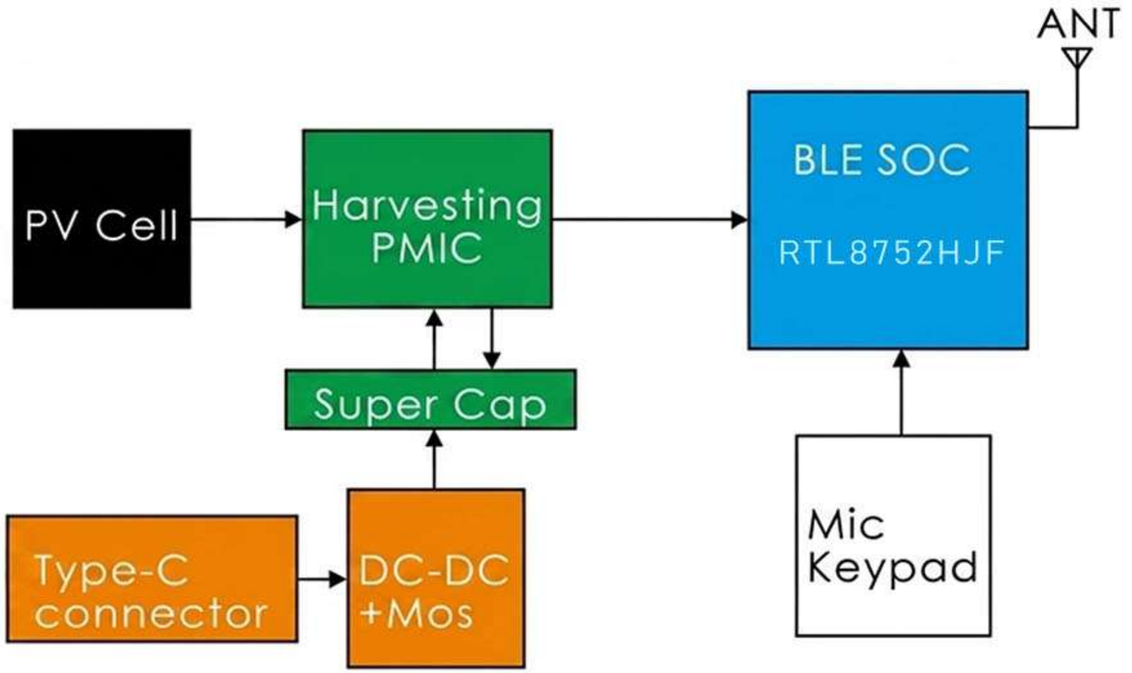
- ✓ Low power BLE SOC
- ✓ Can adapt to various well-known PV cell brands
- ✓ Can efficiently harvests energy in Low-light
- ✓ Delivers the energy required for BLE operation
- ✓ Battery Free(Energy stored in Super Cap)
- ✓ GTV/ATV/AOSP



ATB1113 For MTK mainboard ATV GTV and All AOSP mainboard.

Remote Source's 2 ICs Energy Harvesting Solution 3

- ✓ Low power BLE SOC
- ✓ Can adapt to various well-known PV cell brands
- ✓ Can efficiently harvests energy in Low-light
- ✓ Delivers the energy required for BLE operation
- ✓ Battery Free(Energy stored in Super Cap)
- ✓ GTV/ATV/AOSP



RTL8752HJF For Realtek mainboard ATV GTV and All AOSP mainboard.

Power Consumption and Power Comparison

- ✓ Comparing Remote Source Eco Remote control with Mass Market Android TV Remote Control
- ✓ Measuring power consumption directly from Batteries
- ✓ No energy harvesting

Mode	Remote Source BLE	BLE Remote
Standby (idle)	5.2 µA	48.0 µA
Button Press	0.268 mA	2.86 mA
Voice Command	1.26 mA	20 mA

Almost 10 times of Battery Life



Simulation Results

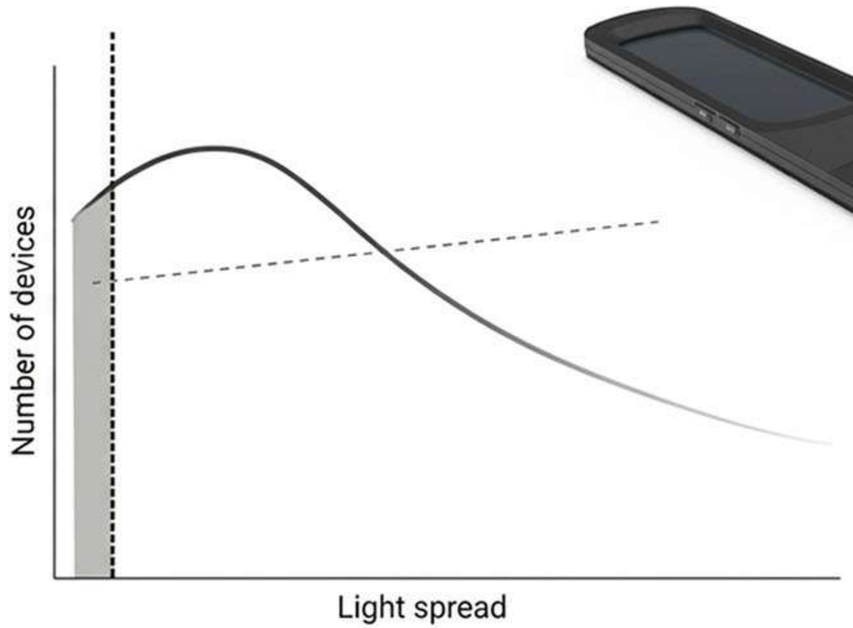
Standard battery, Rechargeable battery or Battery free options can store sufficient energy to meet daily remote usage

	Cell Size (cm^2)	Standard Coin Cell plus Cap	Standard Coin Cell plus Supercap	Rechargeable Battery	Hybrid Supercap
Well Lit Room	10.9	8.1 years	Forever	Forever	Forever
Dimly Lit Room	10.9	5.1 years	Forever	Forever	Forever

Forever indicates daily harvested energy sufficient to meet average daily power consumption

Handling the very low light corner cases.

Creating a Robust System







Message overlay before remote runs out of charge.
+
Opportunity for interaction with consumer with branding and sustainability message.



Optional USB for “heart starting” charge function



Appearance：

- Meets the Google GTV RCU 2024 standard
- Acrylic cover, invisible photovoltaic panel
- Product Dimensions：173.7*38.9*9mm

Technical Specifications：

- Total 23 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery，Charged by solar energy
- 50F supercapacitor
- Solar irradiance area：Hidden type
- Charging time（USB)：15 minutes



Appearance:

- Meets the Google GTV RCU 2024 standard
- Eco-friendly plastic, Novel patterns, visual impact, and full tactile sensation
- Product Dimensions: 178.1*40*13.3mm

Technical Specifications:

- Total 23 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 50F supercapacitor
- Solar irradiance area: Hidden type
- Charging time (USB): 15 minutes



Appearance:

- Meets the Google GTV RCU 2024 standard
- Bifacial Solar Harvesting PV panel available
- Product Dimensions: 166.4*40.4*13.3mm

Technical Specifications:

- Total 23 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 50F supercapacitor
- Solar irradiance area: Hidden type
- Charging time (USB): 15 minutes



Appearance:

- Eco-friendly plastic, Novel patterns, visual impact, and full tactile sensation
- Product Dimensions: 175*41*17mm

Technical Specifications:

- Total 20 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 50F supercapacitor
- Solar irradiance area: Hidden type
- Charging time (USB): 15 minutes



Appearance:

- Eco-friendly plastic, Novel patterns, visual impact, and full tactile sensation
- Product Dimensions：145*39*17mm

Technical Specifications:

- Total 23 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery，Charged by solar energy
- 50F supercapacitor
- Solar irradiance area：Hidden type
- Charging time（USB)：15 minutes



Appearance:

- Eco-friendly plastic, Novel patterns, visual impact, and full tactile sensation
- Product Dimensions: 200*43.6*13.5mm



Technical Specifications:

- Total 24 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 50F supercapacitor
- Solar irradiance area: Hidden type
- Charging time (USB): 15 minutes

Appearance:

- Eco-friendly plastic, Elegant patterns, visual impact, and full tactile sensation
- Product Dimensions: 175*42*16mm

Technical Specifications:

- Total 23 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 50F supercapacitor
- Solar irradiance area: Hidden type
- Charging time (USB): 15 minutes



Appearance:

- Eco-friendly plastic, Elegant patterns, visual impact, and full tactile sensation
- Product Dimensions: 175*42*16mm

Technical Specifications:

- Total 23 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 50F supercapacitor
- Solar irradiance area: Hidden type
- Charging time (USB): 15 minutes



Appearance：

- Eco-friendly plastic, Elegant patterns, visual impact, and full tactile sensation
- Product Dimensions：195*42*16mm

Technical Specifications：

- Total 32 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery，Charged by solar energy
- 50F supercapacitor
- Solar irradiance area：Hidden type
- Charging time（USB)：15 minutes



REMOTE SOURCE

Appearance:

- Eco-friendly plastic, with a frosted surface treatment
- Swing button: VOL+ / VOL-, CH+/CH-
- Product Dimensions: 175*43*13.5mm

Technical Specifications:

- Total 25~27 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery, Charged by solar energy
- 20F supercapacitor
- Solar irradiance area: 33*38mm
- Charging time (USB): 15 minutes



Appearance：

- Various photovoltaic panel Options
- Swing button：VOL+ / VOL-，CH+/CH-
- Product Dimensions： 175*43*13.5mm

Technical Specifications：

- Total 25~27 Keys
- BLE & IR
- Voice Command
- LED indication
- No Battery，Charged by solar energy
- 20F supercapacitor
- Charging time（USB）：15 minutes





REMOTE SOURCE

888-769-8806

inquiries@remotesource.net

www.remotesource.net